

Examination Regulations

During the examination, only the following are allowed:

- Microsoft SQL Server.
- SQL Server Management Studio (SSMS) without any AI features, AI assistants, AI plug-ins, or AI extensions enabled or installed.
- Printed materials or offline documents that are already stored on your computer before the examination.
- A network connection **only** to the **ExamOnline** Wi-Fi network for examination purposes.

The following are strictly prohibited:

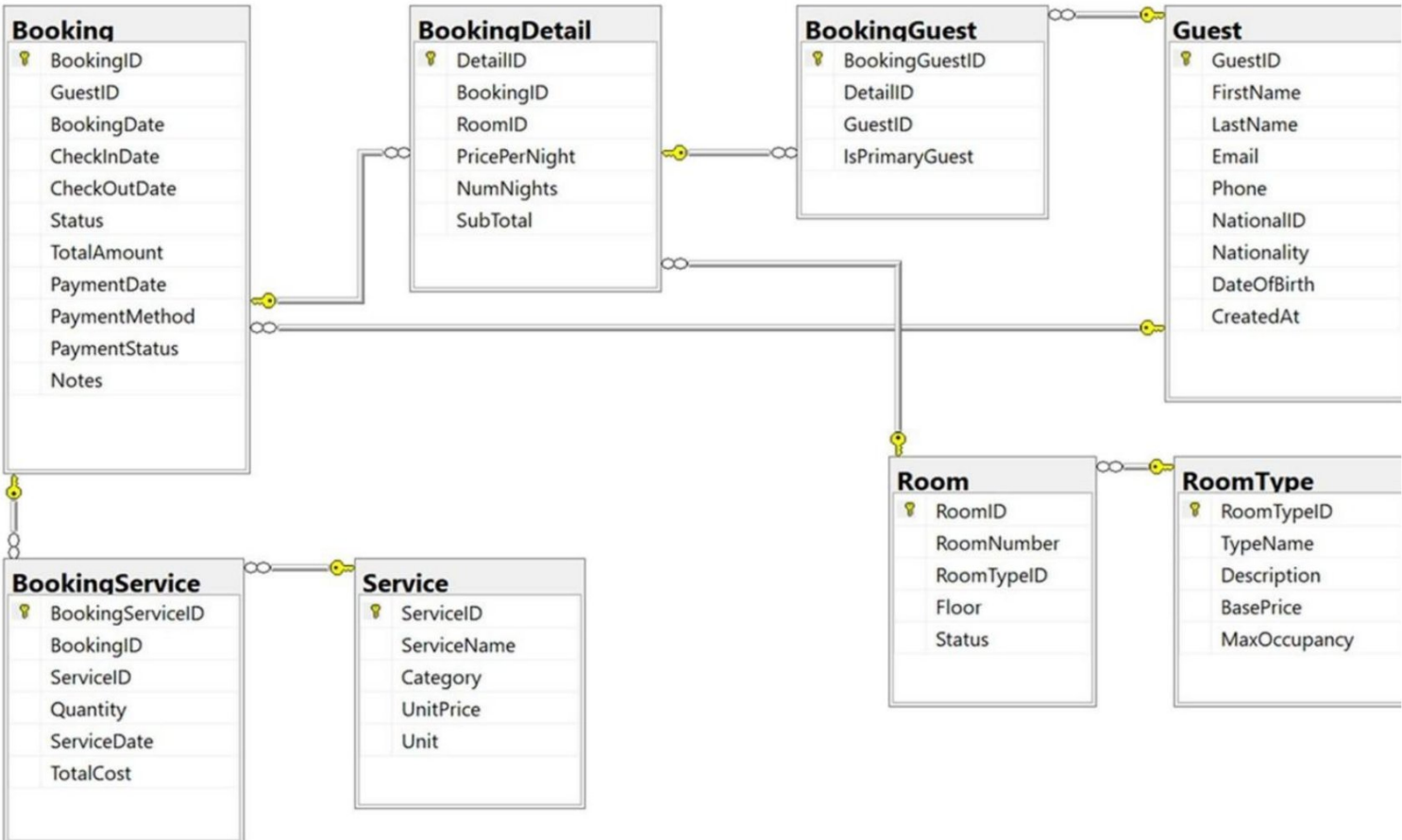
- Connecting to any network other than the **ExamOnline** Wi-Fi network, including personal hotspots, mobile data, VPNs, remote desktop services, cloud storage, or any other Internet connection.
- Accessing any website or online service that is not required for the examination.
- Any online or offline AI tools, including but not limited to ChatGPT, Microsoft Copilot, GitHub Copilot, Gemini, Claude, DeepSeek, Cursor AI, Codeium, Tabnine, Amazon Q, NotebookLM, LMStudio, Visual Studio Code (VSCode), or any other AI coding assistant.
- Any AI-powered extensions, plug-ins, browser add-ons, or IDE/SSMS extensions that provide code generation, code completion, explanations, or suggestions.
- Any communication or file sharing with other candidates or external parties.

Violation Policy

Failure to comply with any of the requirements or regulations listed above will be treated as academic misconduct, and the examination score will be 0.

Database for Question 2 onward

Starting from Question 2, you are required to work with the database supplied in the accompanying .sql file. Before answering the questions, run the provided script so that the required tables are created and populated with the supplied data. The database schema is illustrated in the diagram below.



(Hình ảnh: SSMS database diagram showing 8 tables with their columns, primary keys (key icon) and foreign-key relationship lines (one-to-many symbols). Tables and columns:

- **Booking** (PK BookingID): BookingID, GuestID, BookingDate, CheckInDate, CheckOutDate, Status, TotalAmount, PaymentDate, PaymentMethod, PaymentStatus, Notes.
 - **BookingDetail** (PK DetailID): DetailID, BookingID, RoomID, PricePerNight, NumNights, SubTotal.
 - **BookingGuest** (PK BookingGuestID): BookingGuestID, DetailID, GuestID, IsPrimaryGuest.
 - **Guest** (PK GuestID): GuestID, FirstName, LastName, Email, Phone, NationalID, Nationality, DateOfBirth, CreatedAt.
 - **BookingService** (PK BookingServiceID): BookingServiceID, BookingID, ServiceID, Quantity, ServiceDate, TotalCost.
 - **Service** (PK ServiceID): ServiceID, ServiceName, Category, UnitPrice, Unit.
 - **Room** (PK RoomID): RoomID, RoomNumber, RoomTypeID, Floor, Status.
 - **RoomType** (PK RoomTypeID): RoomTypeID, TypeName, Description, BasePrice, MaxOccupancy.
- Relationship lines (foreign keys) visible in the diagram: Booking.GuestID -> Guest.GuestID; BookingDetail.BookingID -> Booking.BookingID; BookingDetail.RoomID ->

Question 1

Set up a single database, then write SQL statements to create, within that database, all tables derived from the ERD shown in the following picture, with appropriate attributes, primary keys, and foreign keys.

Note that when writing the SQL commands as requested, you MUST keep the names of the tables, relationships, attributes, and attribute data types EXACTLY as provided in the given ERD.

Attributes with underline are Primary Key of each entity.

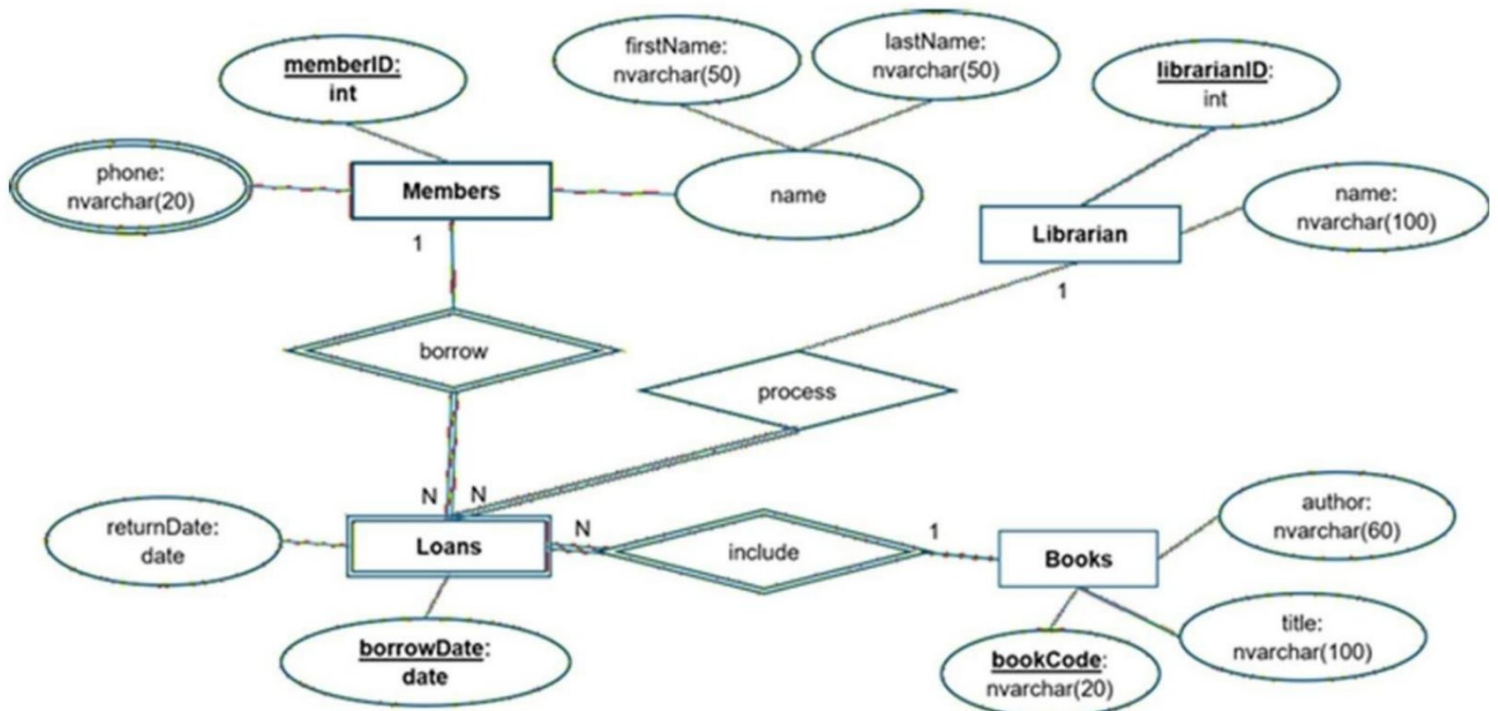
Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.

If a table must be created for representing a relationship, the name of the table must be the same as the name of the relationship.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database_name" or any statements using database's name in your submission.

You must use ER style conversions for subclasses in this diagram if they exist.

The table represents the multi-values attribute must have the name which is the concatenation of the table name and the attribute name. For example, if table named Student has the multi-values attribute named Phone, the name of the corresponding table must be StudentPhone.



(Hình ảnh: An Entity-Relationship diagram (Chen notation) captioned "Picture 1.1" with four entities (rectangles), their attributes (ellipses) and three relationships (diamonds):

- Entity **Members** with attributes: memberID: int (underlined = primary key); phone: nvarchar(20) drawn as a DOUBLE ellipse (multi-valued attribute); and a composite attribute "name" (plain ellipse) composed of firstName: nvarchar(50) and lastName: nvarchar(50).
 - Entity **Librarian** with attributes: librarianID: int (underlined = primary key); name: nvarchar(100).
 - Entity **Loans** drawn as a DOUBLE rectangle (weak entity) with attributes: borrowDate: date (underlined = primary key); returnDate: date.
 - Entity **Books** with attributes: bookCode: nvarchar(20) (underlined = primary key); author: nvarchar(60); title: nvarchar(100).
- Relationships:
- borrow** (drawn as a double diamond / identifying relationship) between Members and Loans, cardinality Members (1) to Loans (N).
 - process** between Librarian and Loans, cardinality Librarian (1) to Loans (N).
 - include** (drawn as a double diamond) between Loans and Books, cardinality Loans (N) to Books (1).]

Question 2

Write an SQL query that returns all information about guests created between March 1, 2024 and May 31, 2024 (inclusive), as displayed in the following figure:

Picture 2.1

GuestID	FirstName	LastName	Email	Phone	NationalID	Nationality	DateOfBirth	CreatedAt
3	Chen	Wei	chen.wei3@email.com	0900000003	NID0003	Chinese	1992-03-27	2024-03-04 00:00:00.000
5	Hiroshi	Tanaka	hiroshi.tanaka5@email.com	0900000005	NID0005	Japanese	1979-05-25	2024-05-08 00:00:00.000
6	Linh	Tran	linh.tran6@email.com	0900000006	NID0006	Vietnamese	1990-12-10	2024-05-02 00:00:00.000
10	Carlos	Rodriguez	carlos.rodriguez10@email.com	0900000010	NID0010	Spanish	1999-04-06	2024-03-24 00:00:00.000
12	Emily	Johnson	emily.johnson12@email.com	0900000012	NID0012	American	1986-06-04	2024-05-28 00:00:00.000
15	Ahmed	Ali	ahmed.ali15@email.com	0900000015	NID0015	Egyptian	1984-03-15	2024-03-17 00:00:00.000
17	Pierre	Dupont	pierre.dupont17@email.com	0900000017	NID0017	French	1971-05-13	2024-05-21 00:00:00.000
20	Tran	Dao	tran.dao20@email.com	0900000020	NID0020	Vietnamese	2004-02-24	2024-03-02 00:00:00.000
22	Anna	Kovacs	anna.kovacs22@email.com	0900000022	NID0022	Hungarian	1991-04-22	2024-05-06 00:00:00.000
27	Omar	Ibrahim	omar.ibrahim27@email.com	0900000027	NID0027	Jordanian	1976-03-03	2024-03-22 00:00:00.000
32	Sakura	Nakamura	sakura.nakamura32@email.com	0900000032	NID0032	Japanese	1996-02-12	2024-03-15 00:00:00.000
34	Amira	Khalil	amira.khalil34@email.com	0900000034	NID0034	Jordanian	1983-04-10	2024-05-19 00:00:00.000
39	Wei	Liu	wei.liu39@email.com	0900000039	NID0039	Chinese	2003-03-19	2024-05-04 00:00:00.000
44	Jordan	Morgan	jordan.morgan44@email.com	0900000044	NID0044	British	1988-02-28	2024-04-03 00:00:00.000
49	Raj	Singh	raj.singh49@email.com	0900000049	NID0049	Indian	1973-01-09	2024-03-13 00:00:00.000

Question 3

Write an SQL query to show the BookingID, GuestName (formed by concatenating FirstName, a single space, and LastName), Nationality, CheckInDate, BookingServiceID, ServiceName, and Category for every service of the 'Food & Beverage' category used in bookings with a CheckInDate in 2025, where GuestName is the name of the guest who made the booking, as shown below:

Picture 3.1

BookingID	GuestName	Nationality	CheckInDate	BookingServiceID	ServiceName	Category
67	Michael Johnson	American	2025-10-24	132	Breakfast Room Service	Food & Beverage
68	Michael Johnson	American	2025-06-02	135	Breakfast Room Service	Food & Beverage
69	Anna Kovacs	Hungarian	2025-01-05	139	Breakfast Room Service	Food & Beverage
77	Astrid Hansen	Norwegian	2025-09-04	152	Breakfast Room Service	Food & Beverage
78	Astrid Hansen	Norwegian	2025-04-07	155	Breakfast Room Service	Food & Beverage
79	Omar Ibrahim	Jordanian	2025-11-10	159	Breakfast Room Service	Food & Beverage
87	Hassan Al-Farsi	Omani	2025-08-09	172	Breakfast Room Service	Food & Beverage
88	Amira Khalil	Jordanian	2025-03-12	175	Breakfast Room Service	Food & Beverage
89	Luke Anderson	Australian	2025-10-15	179	Breakfast Room Service	Food & Beverage
97	Alex Taylor	Australian	2025-06-14	192	Breakfast Room Service	Food & Beverage
98	Jordan Morgan	British	2025-01-17	195	Breakfast Room Service	Food & Beverage
99	Sam Davis	American	2025-08-20	199	Breakfast Room Service	Food & Beverage

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Question 4

Write an SQL query to show RoomTypeID, TypeName, Description, RoomNumber, Floor, DetailID, BookingGuestID, FirstName, LastName, and Nationality for every room type whose TypeName is 'Standard' or 'Suite'.

For every such room type, list its rooms, the booking details for those rooms, and the Japanese guests assigned to each booking detail. The result must include:

- every Standard and Suite room type, even if it has no rooms;
- every room of those types, even if it has no booking details;

In all such cases, the missing columns must appear as NULL.

Order the output by TypeName in descending order, then by DetailID in descending order, and finally by BookingGuestID in ascending order, as shown below:

Picture 4.1

RoomTypeID	TypeName	Description	RoomNumber	Floor	DetailID	BookingGuestID	FirstName	LastName	Nationality
3	Suite	Luxury suite with living area and minibar	602	6	166	351	Kenji	Yamamoto	Japanese
3	Suite	Luxury suite with living area and minibar	602	6	166	352	Sakura	Nakamura	Japanese
3	Suite	Luxury suite with living area and minibar	606	6	140	298	Yuki	Suzuki	Japanese
3	Suite	Luxury suite with living area and minibar	601	6	135	285	Sakura	Nakamura	Japanese
3	Suite	Luxury suite with living area and minibar	605	6	109	229	Hiroshi	Tanaka	Japanese
3	Suite	Luxury suite with living area and minibar	603	6	47	97	Hiroshi	Tanaka	Japanese
3	Suite	Luxury suite with living area and minibar	606	6	20	40	Kenji	Yamamoto	Japanese
3	Suite	Luxury suite with living area and minibar	602	6	16	30	Hiroshi	Tanaka	Japanese
3	Suite	Luxury suite with living area and minibar	604	6	NULL	NULL	NULL	NULL	NULL
1	Standard	Cozy room with basic amenities	101	1	181	389	Kenji	Yamamoto	Japanese
1	Standard	Cozy room with basic amenities	301	3	155	331	Hiroshi	Tanaka	Japanese
1	Standard	Cozy room with basic amenities	301	3	125	267	Yuki	Suzuki	Japanese
1	Standard	Cozy room with basic amenities	101	1	121	259	Yuki	Suzuki	Japanese
1	Standard	Cozy room with basic amenities	201	2	63	134	Sakura	Nakamura	Japanese
1	Standard	Cozy room with basic amenities	102	1	62	133	Hiroshi	Tanaka	Japanese
1	Standard	Cozy room with basic amenities	101	1	61	130	Kenji	Yamamoto	Japanese
1	Standard	Cozy room with basic amenities	301	3	5	9	Kenji	Yamamoto	Japanese
1	Standard	Cozy room with basic amenities	101	1	1	2	Yuki	Suzuki	Japanese
1	Standard	Cozy room with basic amenities	202	2	NULL	NULL	NULL	NULL	NULL
1	Standard	Cozy room with basic amenities	302	3	NULL	NULL	NULL	NULL	NULL

Question 5

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Write a SQL query that generates a report on the performance of each room type for January 2024. The report must include the following columns:

- RoomTypeID - the room type identifier.
- TypeName - the room type name.
- TotalBookings - the total number of distinct bookings that checked in during January 2024.
- TotalRevenue - the total revenue generated from bookings that checked in during January 2024.
- AveragePricePerNight - the average revenue earned per occupied night, calculated as:

TotalRevenue/TotalNumberOfNights, where the total number of nights (TotalNumberOfNights) is the sum of NumNights for all bookings that checked in during January 2024 of the room type.

Requirements:

- Consider only bookings with a CheckInDate in January 2024.
- Include all room types, even if they have no bookings during this period.
- Display 0 for TotalRevenue and AveragePricePerNight when there are no bookings.

- Display AveragePricePerNight as a DECIMAL(10,2) value.
- Sort the result by TotalRevenue in descending order. If two room types have the same revenue, sort them by TypeName in ascending order.

Picture 5.1

RoomTypeID	TypeName	TotalBookings	TotalRevenue	AveragePricePerNight
2	Deluxe	2	4410.00	163.33
4	Family	1	2520.00	210.00
1	Standard	3	1808.00	86.10
5	Penthouse	1	1200.00	600.00
3	Suite	0	0.00	0.00

Question 6

Write a SQL query that generates a report summarizing service usage by service category for the year 2024 (based on ServiceDate). The report must include the following columns:

- Category - the service category.
- NumberOfServices - the total number of services in the category (include all services, even if they were not used in any booking during 2024).
- MinBookingsPerService - the minimum number of bookings in which a service from the category was used during 2024.
- MaxBookingsPerService - the maximum number of bookings in which a service from the category was used during 2024.

Picture 6.1

Category	NumberOfServices	MinBookingsPerService	MaxBookingsPerService
Business	1	12	12
Food & Beverage	3	0	18
Housekeeping	1	24	24
Recreation	2	6	24
Transport	2	6	18
Wellness	1	12	12

Question 7

Write an SQL query that identifies the guest(s) with the highest total revenue for each nationality in 2024. For every nationality, show the guest(s) whose total revenue (computed as the sum of room charges (SubTotal) and service charges (TotalCost)) from all of his/her bookings with a CheckInDate in 2024 is the highest among all guests of the same nationality. Guests who had no bookings in 2024 must also be included, with their TotalRevenue reported as 0.

Each result row must display:

- Nationality
- GuestID
- GuestName (the concatenation of the guest's first name, a single space, and the last name)
- TotalRevenue (room charges + service charges)

If several guests from the same nationality share the same highest TotalRevenue, include all of them in the result.

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Order the output by Nationality in ascending order and then by GuestID in ascending order for guests of the same nationality.

Picture 7.1

Nationality	GuestID	GuestName	TotalRevenue
American	2	Bob Smith	6650.00
Australian	35	Luke Anderson	0.00
Australian	43	Alex Taylor	0.00
British	7	James Wilson	10040.00

Chinese	3	Chen Wei	6289.00
Egyptian	16	Fatima Hassan	9968.00
French	17	Pierre Dupont	7190.00
German	41	Thomas Mueller	0.00
German	42	Claire Wagner	0.00
Ghanaian	50	Aisha Osei	0.00
Hungarian	22	Anna Kovacs	0.00
Indian	23	Ravi Sharma	0.00
Indian	24	Priya Patel	0.00
Indian	49	Raj Singh	0.00
Italian	29	Marco Rossi	0.00
Italian	30	Giulia Ferrari	0.00
Japanese	11	Yuki Suzuki	8448.00
Jordanian	27	Omar Ibrahim	0.00
Jordanian	34	Amira Khalil	0.00
Korean	47	Lee Park	0.00
Korean	48	Kim Choi	0.00
Lebanese	28	Layla Mansour	0.00
Norwegian	26	Astrid Hansen	0.00
Omani	33	Hassan Al-Far...	0.00
Russian	37	Ivan Petrov	0.00
Russian	38	Natasha Ivano...	0.00
Spanish	4	Maria Garcia	17428.00
Swedish	25	Lars Eriksson	0.00
Vietnam...	8	Sophie Pham	4899.00

Question 8

Create a stored procedure named `addService` that adds a service to an existing booking.

The procedure accepts the following input parameters:

- `@bookingID` INT: the first parameter, the booking ID.
- `@serviceID` INT: the second parameter, the service ID.
- `@quantity` INT: the third parameter, the quantity of the service.

The procedure must meet the following requirements:

- Validate the input data:
 - The `bookingID` must exist in the `Booking` table.
 - The `serviceID` must exist in the `Service` table.
 - The quantity must be greater than or equal to 0.
 - If any validation fails:
 - Print the message "Input values are not valid."
 - Do not insert any record into the `BookingService` table.
 - And return 1
 - Otherwise:
 - Insert a new record into the `BookingService` table with the corresponding parameters `@bookingID`, `@serviceID`, `@quantity`. The `TotalCost` must be calculated as `Quantity x UnitPrice` of the corresponding service.
 - And return 0

For example, when the `addService` procedure is executed with the following statements, it returns 1 in the first three scenarios because the input parameters are not valid.

In the last scenario, it returns 0 and a new row is inserted into the `BookingService` table with `bookingID` = 100, `serviceID` = 10, `quantity` = 2, `TotalCost` = 20, and the

ServiceDate is set to its default value (getdate()).

```
-- Scenario 1: status = 1, no row is inserted
declare @status int
exec @status = addService 1000, 5, 2
select @status as Status

-- Scenario 2: status = 1, no row is inserted
declare @status int
exec @status = addService 99, 15, 2
select @status as Status

-- Scenario 3: status = 1, no row is inserted
declare @status int
exec @status = addService 99, 5, -2
select @status as Status

-- Scenario 4: status = 0 and new row is inserted
declare @status int
exec @status = addService 100, 10, 2
select @status as Status

select *
from BookingService
```

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Question 9

Create a SQL Server trigger named trg_PreventRoomOverbooking on the BookingDetail table. The trigger runs when an insert statement is executed to add one or more rows into the BookingDetail table. For every inserted row, the trigger must meet the following requirements:

- Automatically calculate NumNights as the number of days between the corresponding booking's CheckInDate and CheckOutDate. Ignore the value supplied by the user.
- Prevent inserting a booking detail if:
 - The BookingID does not exist in the Booking table.
 - The RoomID does not exist in the Room table.
 - The same RoomID already exists in the same booking.
 - The room has already been assigned to another booking whose Status is not 'Cancelled' and whose booking period (from CheckInDate to CheckOutDate) overlaps with the new booking's stay.
 - If an inserted row violates either rule above, it must not be inserted.
 - Rows that satisfy all requirements should be inserted successfully. Valid rows should still be inserted, even if other rows in the same INSERT statement fail the trigger conditions.

After creating the trigger, execute the following test statement; the result should be as follows:

- The first row must be ignored because the bookingID = 150 does not exist in the Booking table and the roomID = 50 does not exist in the Room table.
- The second row must be ignored because the room with roomID = 1 already exists in the booking with bookingID = 1.
- The third row must be ignored because the room with roomID = 3 already exists in another booking whose booking period overlaps with the booking with bookingID = 19.
- The final row will be inserted but with the NumNights = 2 as in the following figure because the booking with bookingID = 2 stays only 2 nights (not 4 nights).

```
INSERT BookingDetail(BookingID,RoomID, PricePerNight, NumNights)
VALUES (150,50,50,1), -- invalid bookingID and RoomID
(1,1,88,1), -- room already in the booking
(19,3,80, 4), -- overlap checkInDate and CheckOutDate with another booking
(2,1,88,4) -- valid insert data

select * from BookingDetail
```

Picture 9.1

DetailID	BookingID	RoomID	PricePerNight	NumNights	SubTotal
197	98	17	360.00	7	2520.00
198	99	18	300.00	1	300.00
199	100	19	300.00	2	600.00
200	100	20	330.00	2	660.00
201	2	1	88.00	2	176.00

Question 10

Write a SQL statement that updates the TotalAmount column for bookings whose BookingID is from 1 to 10 (inclusive).

The TotalAmount of each booking is calculated as:

- the total room revenue (sum of SubTotal in the BookingDetail table), plus
- the total service charges (sum of TotalCost in the BookingService table).

If a booking has no room details or no services, treat the corresponding total as 0.

After the update, the first 20 rows of the Booking table appear as follows.

Picture 10.1

BookingID	GuestID	BookingDate	CheckInDate	CheckOutDate	Status	TotalAmount	PaymentDate	PaymentMethod	PaymentStatus	No
1	1	2023-12-31 00:00:00.000	2024-01-01	2024-01-02	Checked-Out	176.00	2024-01-02 00:00:00.000	Credit Card	Completed	Ear req
2	1	2024-08-02 00:00:00.000	2024-08-04	2024-08-06	Checked-Out	549.00	2024-08-06 00:00:00.000	Debit Card	Completed	Air nee
3	1	2024-03-04 00:00:00.000	2024-03-07	2024-03-10	Checked-Out	284.00	2024-03-10 00:00:00.000	Bank Transfer	Completed	Ho dec
4	1	2024-10-06 00:00:00.000	2024-10-10	2024-10-14	Confirmed	1328.00	2024-10-14 00:00:00.000	Online	Pending	VIP
5	2	2024-05-08 00:00:00.000	2024-05-13	2024-05-18	Confirmed	2475.00	2024-05-18 00:00:00.000	Cash	Pending	Birt cel
6	2	2024-12-10 00:00:00.000	2024-12-16	2024-12-22	Checked-In	900.00	2024-12-22 00:00:00.000	Credit Card	Completed	Ani sta
7	2	2024-07-12 00:00:00.000	2024-07-19	2024-07-26	Cancelled	2215.00	2024-07-26 00:00:00.000	Debit Card	Refunded	Bus
8	2	2024-02-14 00:00:00.000	2024-02-22	2024-02-23	No-Show	1060.00	2024-02-23 00:00:00.000	Bank Transfer	Failed	Grc
9	3	2024-09-16 00:00:00.000	2024-09-25	2024-09-27	Confirmed	730.00	2024-09-27 00:00:00.000	Online	Pending	Lor dis apt
10	3	2024-03-24 00:00:00.000	2024-04-03	2024-04-06	Checked-Out	1890.00	2024-04-06 00:00:00.000	Cash	Completed	Co up
11	3	2024-10-26 00:00:00.000	2024-11-06	2024-11-10	Checked-Out	0.00	2024-11-10 00:00:00.000	Credit Card	Completed	NU
12	3	2024-05-28 00:00:00.000	2024-06-09	2024-06-14	Checked-Out	0.00	2024-06-14 00:00:00.000	Debit Card	Completed	NU
13	4	2023-12-30 00:00:00.000	2024-01-12	2024-01-18	Checked-Out	0.00	2024-01-18 00:00:00.000	Bank Transfer	Completed	NU
14	4	2024-08-01 00:00:00.000	2024-08-15	2024-08-22	Confirmed	0.00	2024-08-22 00:00:00.000	Online	Pending	NU
15	4	2024-03-17 00:00:00.000	2024-03-18	2024-03-19	Confirmed	0.00	2024-03-19 00:00:00.000	Cash	Pending	NU
16	4	2024-10-19 00:00:00.000	2024-10-21	2024-10-23	Checked-In	0.00	2024-10-23 00:00:00.000	Credit Card	Completed	Ear req
17	5	2024-05-21 00:00:00.000	2024-05-24	2024-05-27	Cancelled	0.00	2024-05-27 00:00:00.000	Debit Card	Refunded	Air nee
18	5	2023-12-29 00:00:00.000	2024-01-02	2024-01-06	No-Show	0.00	2024-01-06 00:00:00.000	Bank Transfer	Failed	Ho dec
19	5	2024-07-31 00:00:00.000	2024-08-05	2024-08-10	Confirmed	0.00	2024-08-10 00:00:00.000	Online	Pending	VIP
20	5	2024-03-02 00:00:00.000	2024-03-08	2024-03-14	Checked-Out	0.00	2024-03-14 00:00:00.000	Cash	Completed	Birt cel

